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1 Quiz 1

Proof. **Step 1:**
By definition of the greedy policy, for any action $a \in A$:

$$Q_\pi(s,a) \leq Q_\pi\big(s, \pi'(s)\big)$$

Step 2:
Expand $Q_\pi\big(s, \pi'(s)\big)$ using the Bellman expectation equation:

$$Q_\pi(s,a) \leq \mathbb{E}_{\pi'}\big[r_t + \gamma Q_\pi(s_{t+1}, a_{t+1}) \bigm| s_t = s, a_t = \pi'(s)\big]$$

Step 3:
Recursively apply the greedy policy inequality to $Q_\pi(s_{t+1}, a_{t+1})$ (i.e., $Q_\pi(s_{t+1}, a_{t+1}) \leq Q_\pi\big(s_{t+1}, \pi'(s_{t+1})\big)$):

$$\begin{aligned}
Q_\pi(s,a) &\leq \mathbb{E}_{\pi'}\big[r_t + \gamma Q_\pi\big(s_{t+1}, \pi'(s_{t+1})\big) \bigm| s_t = s, a_t = \pi'(s)\big] \\
&= \mathbb{E}_{\pi'}\big[r_t + \gamma r_{t+1} + \gamma^2 Q_\pi\big(s_{t+2}, \pi'(s_{t+2})\big) \bigm| s_t = s, a_t = \pi'(s)\big] \\
&\leq \mathbb{E}_{\pi'}\big[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots \bigm| s_t = s, a_t = \pi'(s)\big]
\end{aligned}$$

Step 4:
The infinite sum on the right-hand side is the definition of the action-value function under policy π':

$$\mathbb{E}_{\pi'}\big[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \bigm| s_t = s, a_t = \pi'(s)\big] = Q_{\pi'}\big(s, \pi'(s)\big)$$

Conclusion:
For the greedy action $a = \pi'(s)$: $Q_\pi\big(s, \pi'(s)\big) \leq Q_{\pi'}\big(s, \pi'(s)\big)$.
For any other action $a \neq \pi'(s)$, by Step 1 + Step 4:

$$Q_\pi(s,a) \leq Q_\pi\big(s, \pi'(s)\big) \leq Q_{\pi'}\big(s, \pi'(s)\big)$$

Since $Q_{\pi'}\big(s, \pi'(s)\big) = \max_{a'} Q_{\pi'}(s, a')$, by transitivity, we have $Q_\pi(s,a) \leq \max_{a'} Q_{\pi'}(s,a')$, which implies $Q_{\pi'}(s,a) \geq Q_\pi(s,a)$ for all $s \in S, a \in A$.

$\square$